

Approximating ODE's

$$\frac{dy}{dt} = y' = f(t, y) \quad \text{w/ IVP } y(t_0) = 1$$

$$\text{Stability: } -1 \leq \sigma \leq 1 \Rightarrow |\sigma| = \left| \frac{y_{i+1}}{y_i} \right| \leq 1$$

Step 1 discretize

$$[t, \infty) \rightarrow t = t_0, t_1, \dots, t_n \quad \Delta t = t_1 - t_0 \quad t_i = \Delta t i \quad \text{for } i = 0, 1, \dots, n$$

Step 2 Choose Approximation Method.

* **Forward Euler**: Accuracy $\rightarrow 1^{\text{st}}$ Order, Stability $\rightarrow$ Conditional, Explicit

$$\left. \frac{dy}{dt} \right|_{t_i}^{\text{FD}} = f(t_i, y_i) = \frac{y(t_i + \Delta t) - y(t_i)}{\Delta t} \Rightarrow y(t_i + \Delta t) = y(t_i) + \Delta t f(t_i, y_i) \Rightarrow y_{i+1} = y_i + \Delta t f(t_i, y_i)$$

* **Backward Euler**: Accuracy $\rightarrow 1^{\text{st}}$ Order, Stability $\rightarrow$ Unconditional, Implicit

$$\left. \frac{dy}{dt} \right|_{t_{i+1}}^{\text{BD}} = f(t, y) \Big|_{t_{i+1}} = \frac{y(t_{i+1}) - y(t_i)}{\Delta t} \Rightarrow y(t_{i+1}) = y(t_i) + \Delta t f(t_{i+1}, y_{i+1}) \Rightarrow y_{i+1} = y_i + \Delta t f(t_{i+1}, y_{i+1})$$

* **Trapezoidal**: Accuracy $\rightarrow 2^{\text{nd}}$ Order, Stability $\rightarrow$ Unconditional, Implicit

$$y_{i+1} = y_i + \frac{\Delta x}{2} (f(x_i, y_i) + f(x_{i+1}, y_{i+1}))$$

* **Predictor-Corrector (Heun's - Midpoint)**: Accuracy $\rightarrow 2^{\text{nd}}$ Order, Stability $\rightarrow$ Conditional, Explicit

* Step 1: Predictor (Using Forward Euler): $y_{i+1}^p = y_i + \Delta x f(x_i, y_i)$

* Step 2: Corrector: $y_{i+1} = y_i + \frac{\Delta x}{2} (f(x_i, y_i) + f(x_{i+1}, y_{i+1}^p))$

* Step 1: Predictor (Using Midpoint): $y_{i+\frac{1}{2}}^p = y_i + \frac{\Delta x}{2} f(x_i, y_i)$

* Step 2: Corrector: $y_{i+1} = y_i + \Delta x f(x_{i+\frac{1}{2}}, y_{i+\frac{1}{2}}^p)$

* **Runge-Kutta 4 (RK4)**: Accuracy $\rightarrow 4^{\text{th}}$ Order, Stability $\rightarrow$ Conditional, Explicit

$$y(x+h) = y(x) + \frac{1}{6} (F_1 + 2F_2 + 2F_3 + F_4) \quad \begin{cases} F_1 = h f(x, y) \\ F_2 = h f(x + \frac{h}{2}, y + \frac{F_1}{2}) \\ F_3 = h f(x + \frac{h}{2}, y + F_2) \\ F_4 = h f(x + h, y + F_3) \end{cases}$$

absolute error $\Delta y = |y - y_i|$
relative error $= \frac{\Delta y}{y}$

Errors: Local Truncation Error: $y \rightarrow$ Exact Sol
Global Errors $\rightarrow$ Method Order Accuracy
 $y \rightarrow$ Approx Sol

Numerical Methods for 2nd order ODEs: General Form - $y'' + ay' + by = c$

Step 1: Transform in to system of 1st order ODEs - Let: $y_1 = y$
 $y_2 = y' = y_1'$
 $y_3 = y'' = y_2' = c - ay_2 - by_1$

Step 2: Apply Numerical Approx. Method to System to get recursion formulas.

Step 3: Stability - 2nd Order ODE's: Consider System of Equations - Find Eigenvalues/roots for Characteristic Equation

Boundary Value Problems (using Shooting Method)

Step 1: Guess Value of missing initial conditions

Step 2: Solve ODE as IVP w/ Numerical Method

Step 3: Compare boundary solution to given boundary Value

Step 4: Update Initial Condition and Repeat.

Linear ODEs

Guess 1

Solution 1

Guess 2

Solution 2

Interpolate:

$$m = \frac{\text{Guess 1} - \text{Guess 2}}{\text{Solution 1} - \text{Solution 2}}$$

$$\text{Calculated Guess 3} = \text{Guess 2} + m(\text{Target} - \text{Solution 2})$$

$\hookrightarrow$ Solution 3 = Target

$\uparrow$ Given Boundary Condition

Example 5: Given - $\frac{dy}{dt} = 8(1+t) - 2y$ | $y(0) = 1$ $\Delta t = 0.2$ solve for next time step

* Using Forward Euler: $y_{i+1} = y_i + \Delta t f(t_i, y_i) \Rightarrow y_{i+1} = y_i + \Delta t (8(1+t_i) - 2y_i) \Rightarrow \dots$

$\dots \Rightarrow y_{i+1} = y_i(1 - 2\Delta t) + 8\Delta t(1+t_i) \Rightarrow y_i = y_0(1 - 2\Delta t) + 8\Delta t(1+t_0) \therefore y_i = 2.2$

* Using Backward Euler: $y_{i+1} = y_i + \Delta t f(t_{i+1}, y_{i+1}) \Rightarrow y_{i+1} = y_i + \Delta t (8(1+t_{i+1}) - 2y_{i+1}) \Rightarrow \dots$

$\dots \Rightarrow y_{i+1}(1 + 2\Delta t) = y_i + 8\Delta t(1+t_{i+1}) \Rightarrow y_i = \frac{y_{i+1} + 8\Delta t(1+t_{i+1})}{1 + 2\Delta t} \therefore y_i = 2.09$

* Using Heun's Method: - Step 1 find Predictor (from F.E.) $\Rightarrow y_{i+1}^P \Rightarrow y_i^P = 2.2$

- Step 2: $y_i = y_0 + \frac{\Delta t}{2} (f(t_0, y_0) + f(t_i, y_i^P)) \Rightarrow y_i = 2.12$

Given: $\frac{dy}{dt} = -12y$, Perform Stability Analysis to find Amplification factor, σ , using midpoint

$$y_{i+\frac{1}{2}} = y_i + \frac{\Delta t}{2} f(t_i, y_i) = y_i + \frac{\Delta t}{2} (-12y_i) = y_i(1 - 6\Delta t)$$

$$\Rightarrow y_{i+1} = y_i + \Delta t f(t_{i+\frac{1}{2}}, y_{i+\frac{1}{2}}) = y_i + \Delta t (-12y_i(1 - 6\Delta t)) = y_i(1 - 12\Delta t + 72\Delta t^2)$$

$$\therefore \sigma = \frac{y_{i+1}}{y_i} = 1 - 12\Delta t + 72\Delta t^2$$

* Determine - Condition for Stability for $\Delta t = 0.1 \Rightarrow |\sigma| \leq 1 \Rightarrow |0.52| \leq 1 \Rightarrow$ Stable @ $\Delta t = 0.1$

$$72\Delta t^2 - 12\Delta t + 1 \leq 1 \quad \left\{ \begin{array}{l} 72\Delta t^2 - 12\Delta t + 1 \geq -1 \\ 72\Delta t^2 - 12\Delta t + 2 \geq 0 \end{array} \right.$$

$$\Rightarrow 72\Delta t^2 \leq 12\Delta t$$

$$\Rightarrow \Delta t \leq \frac{1}{6}$$

$$\therefore 0 \leq \Delta t \leq \frac{1}{6}$$

$$\Rightarrow \frac{12 \pm \sqrt{144 - 576}}{144} \Rightarrow \frac{1 \pm \sqrt{3}i}{12} \text{ imaginary roots}$$

conditionally stable

Given: $y'' + 7y' + 12y = 0$ Find Recursive Formula for Numerically Solving ODE w/ B.E.
Sys:

$$\begin{cases} y = y_1 \\ y' = y_2 = f_1(t, y) \\ y'' = -7y_2 - 12y_1 = f_2(t, y) \end{cases}$$

B.E. method Applied to System

$$\{ y_{1,i+1} = y_{1,i} + \Delta t y_{2,i+1} \}$$

$$y_{2,i+1} = y_{2,i} + \Delta t (-7y_{2,i} - 12y_{1,i})$$

$$y_{2,i+1}(1 + 7\Delta t) = y_{2,i} - 12y_{1,i}\Delta t$$

$$y_{2,i+1}(1 + 7\Delta t) = y_{2,i} - 12(y_{1,i} + \Delta t y_{2,i})\Delta t$$

$$y_{2,i+1}(1 + 7\Delta t + 12\Delta t^2) = y_{2,i} - 12y_{1,i}\Delta t$$

$$\therefore \{ y_{2,i+1} = \frac{y_{2,i} - 12y_{1,i}\Delta t}{1 + 7\Delta t + 12\Delta t^2} \}$$

$$\therefore y_{1,i+1} = y_{1,i} + \Delta t \left(\frac{y_{2,i} - 12y_{1,i}\Delta t}{1 + 7\Delta t + 12\Delta t^2} \right)$$

Given: BVP - $y'' + 9y = 0$ | $y(0) = 1$ $y(1.5) = 4$ Solve w/ Shooting Method to find Initial Condition $y'(0)$ such that step size $\Delta x = 0.5$, use Forward Euler.

Step 1: Linearize Sys:

$$\begin{cases} y = y_1 \\ y' = y_2 = f_1(x, y) \\ y'' = -9y = f_2(x, y) \end{cases}$$

Step 2: Forward Euler for system:

$$\begin{cases} y_{1,i+1} = y_{1,i} + \Delta x f_1(x_i, y_i) = y_{1,i} + y_{2,i} \Delta x \\ y_{2,i+1} = y_{2,i} + \Delta x f_2(x_i, y_i) = y_{2,i} - 9y_{1,i} \Delta x \end{cases} \quad \Delta x = 0.5$$

$$m = \frac{1 - 7}{-5.575 - (-3.125)} = 2.76 = \frac{24}{9}$$

$$\Rightarrow \text{guess } 3 = 7 + \frac{24}{9}(4 + 3.125) = 26$$

$$y_{1,2} = 1 + 0.5(26) = 14$$

$$y_{2,2} = 26 + 0.5(-9(1)) = 21.5$$

$$y_{1,3} = 14 + 0.5(21.5) = 24.75$$

$$y_{2,3} = 21.5 + 0.5(-9(14)) = -41.5$$

$$y_{1,4} = 24.75 + 0.5(-41.5) = 4$$

$$\therefore y(1.5) = 4 \checkmark \therefore y(0) = 26$$

Step 3: Guess $y(0) = 1$

$$y_{1,2} = 1 + 0.5(1) = 1.5$$

$$y_{2,2} = 1 + 0.5(-9(1)) = -3.5$$

$$y_{1,3} = 1.5 + 0.5(-3.5) = -0.25$$

$$y_{2,3} = -3.5 + 0.5(-9(1.5)) = -10.25$$

$$y_{1,4} = -0.25 + 0.5(-10.25) = -5.375$$

$$\therefore y(1.5) = 4 \neq -5.375$$

Guess $y(0) = 7$

$$y_{1,2} = 1 + 0.5(7) = 4.5$$

$$y_{2,2} = 7 + 0.5(-9(1)) = 2.5$$

$$y_{1,3} = 4.5 + 0.5(2.5) = 5.75$$

$$y_{2,3} = 2.5 + 0.5(-9(4.5)) = -17.75$$

$$y_{1,4} = 5.75 + 0.5(-17.75) = -3.125$$

$$\therefore y(1.5) = 4 \neq -3.125$$

